



Adam Optimizer

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Disadvantages of Gradient Descent

Slow Convergence: It can take many iterations to reach the optimal solution, especially if the learning rate is too small.

Sensitive to Learning Rate: Choosing the right learning rate is crucial—too high can cause divergence, while too low can make convergence slow.

Local Minima & Saddle Points: In non-convex functions, gradient descent can get stuck in local minima or saddle points, preventing it from finding the global optimum.

Computationally Expensive: For large datasets, computing gradients for every iteration can be resource-intensive.

Requires Differentiability: It relies on the function being differentiable, making it unsuitable for problems with discontinuous cost functions.

Concept of Moving Averages

A moving average (MA) is a statistical technique used to analyze data points by creating a series of averages of different selections from a full data set.

$$SMA_t = \frac{1}{k} \sum_{i=0}^{k-1} x_{t-i}$$

1) Simple Moving Average (SMA):

A simple moving average is a way to smooth out data by calculating the average of a fixed number of past values (called the window size).

2) Exponential Moving Average (EMA):

Unlike Simple Moving Average (SMA), which gives equal weight to all past data, EMA gives more weight to recent values and less to older ones, or vice versa.

$$EMA_t = \beta \cdot EMA_{t-1} + (1-\beta) \cdot x_t$$

$\beta \in [0,1)$ = smoothing factor

The smoothing factor decides if we should give more value to recent values or old values. Values like 0.9 give more weight to old values and value like 0.8 give more weight to recent one

Simple Moving Average (SMA)

Why SMA ?

1. It smooth out the curve ignoring the noise
2. Generalize trends in data

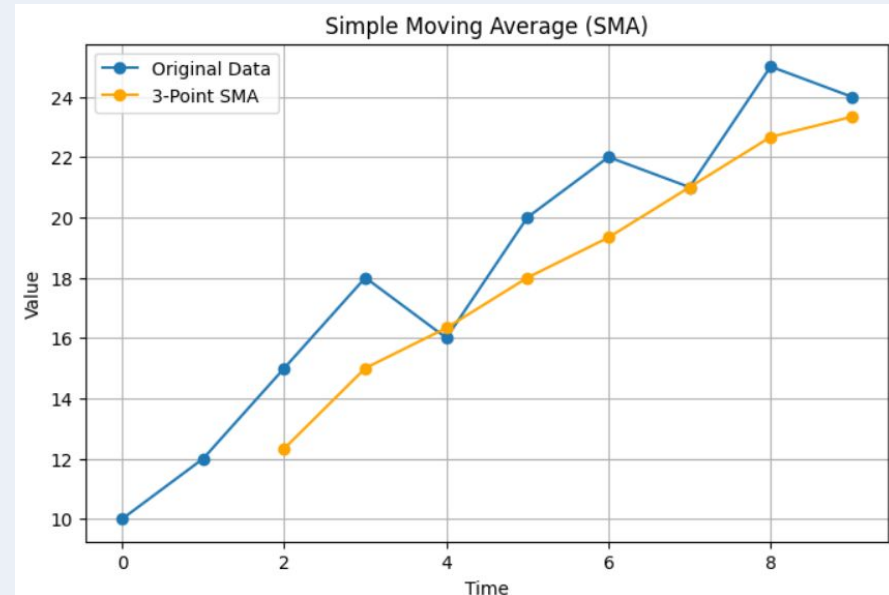
Problem : From the Data beside, calculate the 3 point SMA

steps	value	SMA
0	10	
1	12	
2	15	12.3
3	18	15
4	16	16.3
5	20	18
6	22	19.3
7	21	21
8	25	22.6
9	24	23.3

$$SMA_0 = \text{NULL}$$

$$SMA_1 = \text{NULL}$$

$$SMA_2 = (10 + 12 + 15) / 3 = 12.333$$



Exponential Moving Average (EMA)

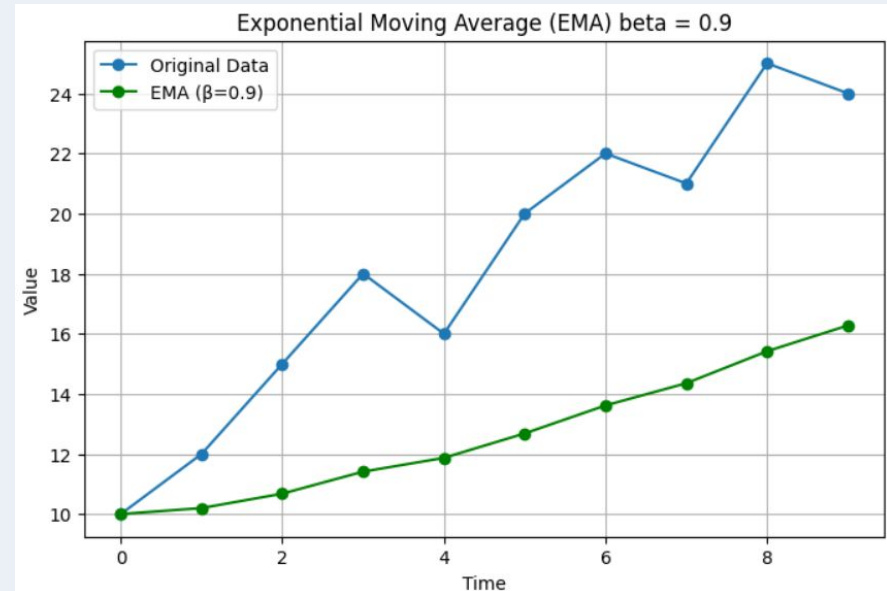
Problem: From the Data beside, calculate the EMA when $\beta = 0.9$

steps	value	EMA
0	10	10
1	12	10.2
2	15	10.7
3	18	11.4
4	16	11.8
5	20	12.6
6	22	13.6
7	21	14.3
8	25	15.4
9	24	16.2

$$\text{EMA}_0 = 10$$

$$\text{EMA}_1 = 0.9 * 10 + 0.1 * 12 = 10.2$$

$$\text{EMA}_2 = 0.9 * 10.2 + 0.1 * 15 = 10.68$$



Concept of Momentum

Why do we need Momentum?

While using GD in bumpy or steep valley cost surfaces. There is a very high probability that GD will:

1. Oscillate back and forth.
2. Move Slowly along Narrow Direction

In such scenarios, momentum helps by accumulating the past gradients, giving the optimizer a push in the right direction—like inertia.

How does Momentum work?

At each step:

1. Compute the gradient:
$$\mathbf{g}_t = \nabla_{\Theta} J(\Theta_t)$$
2. Update the momentum (exponential weighted average of gradients):
$$\mathbf{m}_t = \beta \cdot \mathbf{m}_{t-1} + (1 - \beta) \mathbf{g}_t$$
3. Update the parameters:
$$\Theta_{t+1} = \Theta_t - \alpha \cdot \mathbf{m}_t$$

Terms:

$\mathbf{g}_t$ = Current Gradient

α = Learning rate

$\mathbf{m}_t$ = Accumulated gradient

β = Momentum decay rate (usually 0.9)

Concept of RMSProp

RMS stands for root mean square propagation. It improves optimization by adjusting the learning rate for each parameter.

Why do we need RMSProp?

1. It handles exploding and vanishing gradients
2. Adaptive LR
3. It normalizes updates and makes convergence smooth

How does RMS prop work?

At each step:

1. Compute the gradient:

$$\mathbf{g}_t = \nabla_{\Theta} J(\Theta_t)$$

2. Update the second moment (exponential weighted average of gradients):

$$\mathbf{v}_t = \beta \cdot \mathbf{v}_{t-1} + (1 - \beta) \mathbf{g}_t^2$$

3. Update the parameters:

$$\theta_{t+1} = \theta_t - \frac{\alpha}{\sqrt{\mathbf{v}_t} + \epsilon} \mathbf{g}_t$$

Terms:

$\mathbf{g}_t$ = Current Gradient

α = Learning rate

$\mathbf{v}_t$ = Accumulated squared gradient

β = Momentum decay rate (usually 0.9)

Adam (Adaptive Moment Estimation)

is an optimization algorithm that combines the benefits of Momentum and RMSprop to improve gradient-based learning.

How does RMS prop work?

At each step:

1. Compute the moment component:

$$\mathbf{m}_t = \beta \cdot \mathbf{m}_{t-1} + (1 - \beta) \mathbf{g}_t$$

2. Compute the RMSProp component:

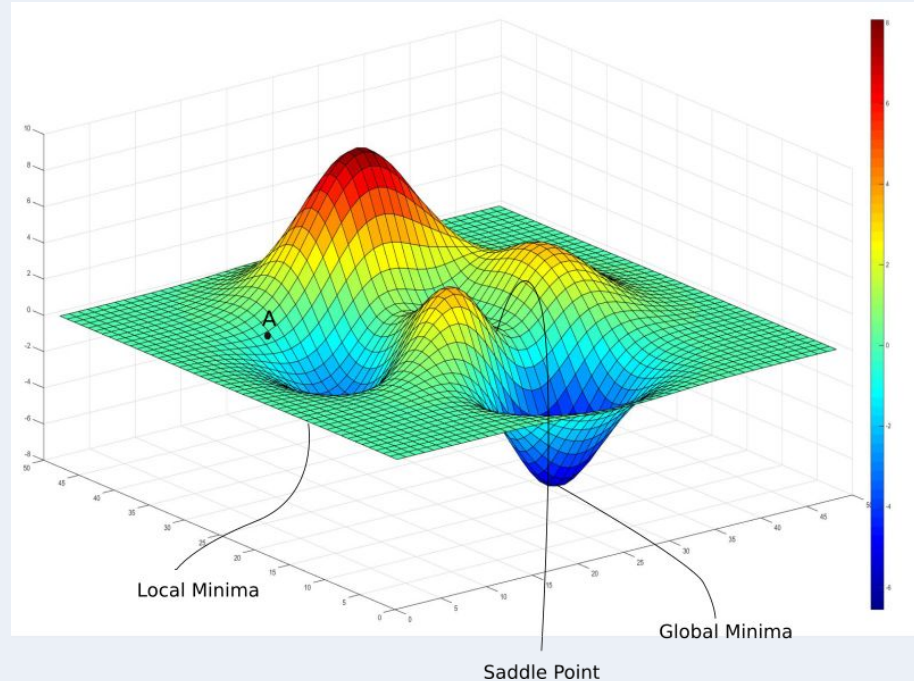
$$\mathbf{v}_t = \beta \cdot \mathbf{v}_{t-1} + (1 - \beta) \mathbf{g}_t^2$$

3. Bias Correction:

$$\hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1 - \beta_1^t}, \quad \hat{\mathbf{v}}_t = \frac{\mathbf{v}_t}{1 - \beta_2^t}$$

4. Parameter Update

$$\theta_{t+1} = \theta_t - \frac{\alpha}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon} \hat{\mathbf{m}}_t$$



The image features a light blue background with abstract, organic shapes in various shades of blue (light blue, medium blue, and dark blue) located in the top-right and bottom-left corners. The text is centered in the middle of the slide.

**THANK YOU
FOR YOUR
ATTENTION!**